

* From Canonical to Configuration space path integral:

$$\begin{aligned} \langle q_f | \hat{U}(t_f, t_i) | q_i \rangle &= \int_{q(t_i)}^{q(t_f)} Dq(t) \int Dp(t) e^{i \int_{t_i}^{t_f} dt L(q, p)} \\ &= \lim_{N \rightarrow \infty} \frac{1}{N!} \int \prod_{n=1}^{N-1} dq_n \prod_{n=0}^{N-1} \int \frac{d^D p_n}{(2\pi\hbar)^D} e^{i \sum_n (p_n(q_{n+1} - q_n) - H(q_n, p_n) \Delta t)} \end{aligned}$$

* Non-rel. particle:

$$\begin{aligned} H(q_n, p_n) &= \sum_{i=1}^D \frac{p_{ni}^2}{2m} + V(q_n) \\ \int \frac{d^D p_n}{(2\pi\hbar)^D} e^{i \sum_n (p_n(q_{n+1} - q_n) - \left(\sum_{i=1}^D \frac{p_{ni}^2}{2m} + V(q_n) \right) \Delta t)} \\ &= e^{i \Delta t \sum_n V(q_n)} \frac{1}{(2\pi\hbar)^D} \left(\frac{\pi 2m}{\Delta t} \right)^{D/2} e^{i \frac{(q_{n+1} - q_n)^2}{2 \left(\frac{\Delta t}{2m} \right)}} \\ &= e^{-i \Delta t \sum_n V(q_n)} \left(\sqrt{\frac{2m\pi}{\Delta t}} \frac{1}{2\pi} \right)^D e^{i \frac{m \Delta t}{2} \left(\frac{q_{n+1} - q_n}{\Delta t} \right)^2} \\ &\approx e^{-i \Delta t \sum_n V(q_n)} \left(\frac{m}{2\pi \Delta t} \right)^{D/2} e^{i \frac{m \Delta t}{2} \dot{q}_n^2} \end{aligned}$$

$$\begin{aligned} \langle q_f | \hat{U}(t_f, t_i) | q_i \rangle &= \lim_{N \rightarrow \infty} \frac{1}{N!} \int \prod_{n=1}^{N-1} dq_n \left(\prod_{n=0}^{N-1} e^{-\Delta t \sum_n V(q_n)} \right) e^{i \frac{m \Delta t}{2} \dot{q}_n^2} \\ &= \left(\lim_{N \rightarrow \infty} \left(\frac{m}{2\pi \Delta t} \right)^{DN/2} \right) \lim_{N \rightarrow \infty} \frac{1}{N!} \int \prod_{n=1}^{N-1} dq_n e^{-\Delta t \sum_n V(q_n)} e^{i \frac{m \Delta t}{2} \dot{q}_n^2} \\ &= C \left(\lim_{N \rightarrow \infty} \prod_{n=1}^{N-1} \int dq_n \right) e^{i \int dt \left(\frac{1}{2} m \dot{q}^2 - V(q) \right)} \\ L &= \frac{1}{2} m \dot{q}^2 - V(q) \end{aligned}$$

$$\langle q_f | \hat{U}(t_f, t_i) | q_i \rangle = C \int_{q(t_i)}^{q(t_f)} Dq(t) e^{i \int_{t_i}^{t_f} dt L(q, \dot{q})}$$

* Work in Fourier space:

$$D = 1 \quad L = \frac{1}{2} m \dot{q}^2 - \frac{1}{2} m \omega^2 q^2$$

q : Solⁿ of classical E.O.M:

$$q(t_f) = q_f, \quad q(t_i) = q_i$$

$$\ddot{q}_{cl} = -\omega^2 q_{cl}$$

$$\left[\frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0 \right.$$

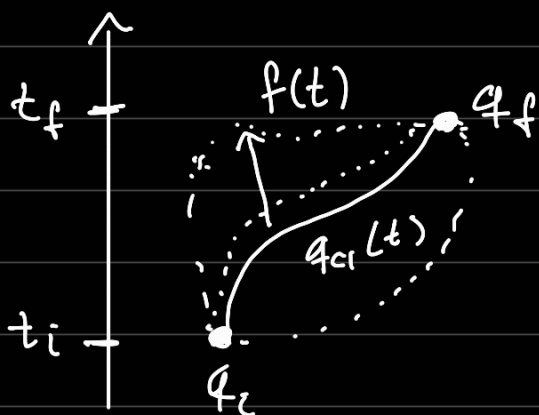
$$\frac{d}{dt} (m \dot{q}) + m \omega^2 q = 0$$

$$\ddot{q}_{cl} + \omega^2 q_{cl} = 0$$

q_{cl} : q which satisfies the classical e.o.m.

$$\boxed{q_{cl}(t_f) = q_f}, \quad \boxed{q_{cl}(t_i) = q_i}$$

$$q(t) = q_{cl}(t) + f(t)$$



→ This is a Homotopy

→ S.M. Path integral → Fundamental Group?

$$\begin{aligned} S[q(t)] &= \int dt \left(\frac{1}{2} m (\dot{q}_{cl} + \dot{f})^2 - \frac{1}{2} m \omega^2 (q_{cl} + f)^2 \right) \\ &= \int dt \left(\frac{1}{2} m (\dot{q}_{cl}^2 + 2 \dot{f} \dot{q}_{cl} + \dot{f}^2) - \frac{1}{2} m \omega^2 (q_{cl}^2 + 2 f q_{cl} + f^2) \right) \\ &= \int dt \left(\frac{1}{2} m \dot{q}_{cl}^2 - \frac{1}{2} m \omega^2 q_{cl}^2 \right) \\ &\quad + \int dt \left(\frac{1}{2} m \dot{f}^2 - \frac{1}{2} m \omega^2 f^2 \right) \\ &\quad + \int dt (m \dot{f} \dot{q}_{cl} - m \omega^2 f q_{cl}) \end{aligned}$$

$$\begin{aligned} &= S[q_{cl}(t)] + S[f(t)] + \left[m f(t) \dot{q}_{cl} \right]_{t_i}^{t_f} - \int_{t_i}^{t_f} dt m \dot{f} \dot{q}_{cl} \\ &\quad - \int_{t_i}^{t_f} dt m \omega^2 f q_{cl} \end{aligned}$$

From B.C.

$0 \leq t \leq t_f$

$$f(t_i) = f(t_f) = 0 \Rightarrow f|_{t_i} = 0$$

$$= S[q_c(t)] + \int dt \left(\frac{1}{2} m \dot{f}^2 - \frac{1}{2} m \omega^2 f^2 \right)$$

* Q: S.M.



QM

Closed - Homotopic curve

"open" - Homotopic

• What does Suss add to homotopic curves?

$$\int_{q(t_i)=q_i}^{q(t_f)=q_f} Dq(t) = \int_{f(t_i)=0}^{f(t_f)=0} Df(t)$$

$$q_n = q(t_n) = q_c(t_n) + \underbrace{f(t_n)}_{f_n}$$

$$dq_n = dq_c + df_n$$

$$\text{constant} \Rightarrow dq_c = 0$$

$$dq_n = df_n$$

$$\langle q_f | \hat{U}(t_f, t_i) | q_i \rangle = C e^{S[q_c]} \int_{f_i}^{f_f} Df(t) e^{i \int dt \left(\frac{1}{2} m \dot{f}^2 - \frac{1}{2} m \omega^2 f^2 \right)}$$

$$* \ddot{q}_{ce} = -\omega q_{ce}$$

$$q_{ce}(t) = q_i \cos(\omega(t - t_i)) + B \sin(\omega(t - t_i))$$

$$q_{ce}(t_f) = q_i \cos(\omega(t_f - t_i)) + B \sin(\omega T) \\ = q_i \cos(\omega T) + B \sin(\omega T)$$

$$\boxed{\frac{q_{ce}(t_f) - q_i \cos(\omega T)}{\sin(\omega T)} = B}$$

$$S[q_{ce}] = \int_{t_i}^{t_f} dt \left(\frac{1}{2} m \dot{q}_{ce}^2 - \frac{1}{2} m \omega^2 q^2 \right)$$

$$\frac{1}{2} m \left[\dot{q} + \ddot{q}^2 = \frac{1}{2} m \left[\dot{q} \ddot{q} \right]^{t_f} - \left[q \ddot{q} \right] \right]$$

$$= \frac{1}{2} m [\dot{q} \dot{q}]_{t_i}^{t_f} + \int \frac{1}{2} m \omega^2 q_{ce}^2$$

$$\therefore S[q_{ce}] = \frac{1}{2} m [\dot{q}_{ce} \dot{q}_{ce}]_{t_i}^{t_f}$$

$$\dot{q}_{ce} = -q_i \omega \sin(\omega(t-t_i)) + \omega B \cos(\omega(t-t_i))$$

$$\dot{q}_{ce}|_i = \omega B \quad |(\dot{q}_{ce})_f = -q_f \omega \sin(\omega T) + \omega B \cos(\omega T)$$

$$S[q_{ce}] = \frac{1}{2} m \left(-q_f q_i \omega \sin(\omega T) + \omega B q_f \cos(\omega T) - q_i \omega B \right)$$

$$S[q_{ce}] = \frac{1}{2} m \left(\omega B q_f \cos(\omega T) - q_f q_i \omega \sin(\omega T) - q_i \omega B \right)$$

$$\bullet - q_i \omega \left(\frac{q_f - q_i \cos(\omega T)}{\sin(\omega T)} \right) = \frac{-q_i q_f \omega + q_i^2 \omega \cos(\omega T)}{\sin(\omega T)}$$

$$\bullet \frac{\omega q_f \cos(\omega T) \left(\frac{q_f - q_i \cos(\omega T)}{\sin(\omega T)} \right)}{\sin(\omega T)} = \frac{\omega q_f^2 \cos(\omega T) - \omega q_f q_i \cos^2(\omega T)}{\sin(\omega T)}$$

$$\Rightarrow \frac{1}{\sin(\omega T)} \left(\omega q_f^2 \cos(\omega T) - \omega q_i q_f \cos^2 - \omega q_i q_f \sin^2 - q_i q_f \omega + q_i^2 \omega \cos(\omega T) \right)$$

$$S[q_{ce}] = \frac{m}{2 \sin(\omega T)} \left(\omega (q_f^2 + q_i^2) \cos(\omega T) - 2 q_i q_f \omega \right)$$

* In practice, we can ignore $\int Df$ as a const. but we'll do this explicitly:

• Recall

$$q(t) = q_{cl}(t) + f(t)$$

$$q(t_i) = q_{cl}(t_i), \quad \begin{cases} f(t_i) = 0 \\ \sum f(t_f) = 0 \end{cases}$$

* Fourier expand $f(t)$

$$f(t) = \sum_{p=1}^{\infty} \tilde{f}_p \sqrt{\frac{2}{T}} \sin\left(\frac{p\pi}{T}(t - t_i)\right)$$

Discretize: $\int_{f(t_i)=0}^{f(t_f)=0} Df(t) = \left| \frac{1}{T} \right| \int_{n=1}^{N-1} df_n$

$$f_n = \sum_{p=1}^{N-1} \tilde{f}_p \sqrt{\frac{2}{T}} \sin\left(\frac{p\pi}{N} \Delta t\right)$$

$$\Delta t = \frac{T}{N}$$

$$\frac{1}{T} \int_{n=1}^{N-1} df_n = \lim_{N \rightarrow \infty} \frac{N-1}{T} \int_{-\infty}^{\infty} \left| \frac{\partial f_n}{\partial \tilde{f}_p} \right| d\tilde{f}_p \delta_{pn}$$

$$\frac{\partial f_n}{\partial \tilde{f}_p} = \sqrt{\frac{2}{T}} \left(\sum_{p=1}^{N-1} \sin\left(\frac{p\pi}{N}\right) \right) \delta_{np}$$

$$\det \left(\frac{\partial f_n}{\partial \tilde{f}_p} \right) = \left(\frac{2}{T} \right)^{N-1/2} C'' \delta_{np}$$

$$\int_{\tilde{f}_c=0}^{\tilde{f}_f=0} Df = \lim_{N \rightarrow \infty} \frac{N-1}{T} \left(\frac{2}{T} \right)^{\frac{N-1}{2}} C'' \int_{-\infty}^{\infty} d\tilde{f}_p$$

$$\int \dot{f}^2 dt = \int \sum_{p,k} \tilde{f}_p \tilde{f}_k \left(\frac{2}{T} \right) \left(\frac{p\pi}{T} \right) \left(\frac{k\pi}{T} \right) \cos\left(\frac{p\pi}{T}t\right) \cos\left(\frac{k\pi}{T}t\right) dt$$

orthogonal:

$$\int dt \cos\left(\frac{p\pi}{T}t\right) \cos\left(\frac{k\pi}{T}t\right)$$

$$\tilde{t} = \frac{\pi}{T} t \Rightarrow \frac{T}{\pi} d\tilde{t} = dt$$

$$\frac{1}{\pi} \int d\tilde{t} \cos(p\tilde{t}) \cos(k\tilde{t}) = \frac{1}{\pi} \frac{\pi}{2} \delta_{pk}$$

→ We assumed convergence ... $\sum \int \leftrightarrow \int \sum$

$$= \sum_p \hat{f}_p^2 \left(\frac{2}{T}\right) \left(\frac{p\pi}{T}\right)^2 \cdot T \frac{1}{2}$$

$$\therefore \int \frac{1}{2} m \hat{f}^2 dt = \frac{\pi^2 m}{2T} \left[\sum_p \hat{f}_p^2 p^2 \right]$$

$$= \frac{1}{2} m \left(\sum_p \hat{f}_p \frac{p^2 \pi^2}{T} \right)$$

$$\text{and } \int -\frac{1}{2} m \omega^2 f^2 dt = -\frac{1}{2} m \omega^2 \sum_p \hat{f}_p^2$$

$$S[f] = \frac{1}{2} m \left(\sum_{p=1}^{N-1} \hat{f}_p^2 \left(\frac{p^2 \pi^2}{T^2} - \omega^2 \right) \right)$$

$$\int_{-\infty}^{\infty} d\hat{f}_p e^{i \frac{1}{2} m \hat{f}_p^2 \left(\frac{p^2 \pi^2}{T^2} - \omega^2 \right)}$$

$$= \sqrt{\frac{2\pi i}{m \left(\frac{p^2 \pi^2}{T^2} - \omega^2 \right)}}$$

$$\langle q_f | U(t_f, t_i) | q_i \rangle = \lim_{N \rightarrow \infty} e^{i S[q_{cl}]} \left(\frac{mN}{2\pi i T} \right)^{N/2} \left(\frac{2}{T} \right)^{\frac{N-1}{2}}$$

$$\times C'' \times \prod_{p=1}^{N-1} \left(\frac{2\pi i}{m \left(\frac{p^2 \pi^2}{T^2} - \omega^2 \right)} \right)^{1/2} e^{i S[q_{cl}]}$$

$$= \sqrt{\frac{m}{2\pi i}} C''' \lim_{N \rightarrow \infty} \left(\frac{1}{\prod_{p=1}^{N-1} \left(1 - \frac{\omega^2 T^2}{p^2 \pi^2} \right)} \right)^{1/2} e^{i S[q_{cl}]}$$

$$\prod_{p=1}^{\infty} \left(1 - \frac{\omega^2 T^2}{p^2 \pi^2} \right) = \frac{\sin(\omega T)}{\omega T}$$

$$\langle q_f | U(t_f, t_i) | q_i \rangle = \sqrt{\frac{m\omega}{2\pi i \sin(\omega T)}} C''' e^{i S[q_{cl}]}$$

* $T \rightarrow 0$

$$\langle q_f | q_i \rangle = \delta(q_f - q_i)$$

* Correlation fns: